



Second Revised

Generation Interconnection

System Impact Study Report

for

Queue Project AE2-025

LARRABEE 230 KV II

78.36 MW Capacity / 445.2 MW Energy

March 2022

Revision 2

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1 Introduction

This System Impact Study has been prepared in accordance with the PJM Open Access Transmission Tariff, 205, as well as the System Impact Study Agreement between **Atlantic Shores Offshore Wind Project 2, LLC**, the Interconnection Customer (IC), and PJM Interconnection, LLC (PJM), Transmission Provider (TP). The Interconnected Transmission Owner (ITO) is Jersey Central Power and Light Company (JCPL).

2 Preface

The intent of the System Impact Study is to determine a plan, with approximate cost and construction time estimates, to connect the subject generation interconnection project to the PJM network at a location specified by the Interconnection Customer. As a requirement for interconnection, the Interconnection Customer may be responsible for the cost of constructing: Network Upgrades, which are facility additions, or upgrades to existing facilities, that are needed to maintain the reliability of the PJM system. All facilities required for interconnection of a generation interconnection project must be designed to meet the technical specifications (on PJM web site) for the appropriate transmission owner.

In some instances an Interconnection Customer may not be responsible for 100% of the identified network upgrade cost because other transmission network uses, e.g. another generation interconnection or merchant transmission upgrade, may also contribute to the need for the same network reinforcement. The possibility of sharing the reinforcement costs with other projects may be identified in the Feasibility Study, but the actual allocation will be deferred until the System Impact Study is performed.

The System Impact Study estimates do not include the feasibility, cost, or time required to obtain property rights and permits for construction of the required facilities. The project developer is responsible for the right of way, real estate, and construction permit issues. For properties currently owned by Transmission Owners, the costs may be included in the study.

The Interconnection Customer seeking to interconnect a wind or solar generation facility shall maintain meteorological data facilities as well as provide that meteorological data which is required per Schedule H to the Interconnection Service Agreement and Section 8 of Manual 14D.

An Interconnection Customer with a proposed new Customer Facility that has a Maximum Facility Output equal to or greater than 100 MW shall install and maintain, at its expense, phasor measurement units (PMUs). See Section 8.5.3 of Appendix 2 to the Interconnection Service Agreement as well as section 4.3 of PJM Manual 14D for additional information.

3 Revision to May 2021 System Impact Study Report

The AE2-025 System Impact Study report has been updated to reflect retooled load flow and stability analysis. The physical connection costs have not been modified in this revision. Final scope, cost and schedule for the physical interconnection will be provided in the Facilities Study Report.

- **Load Flow:** Retooled analysis shows that the following system reinforcements are required for this project: b3042, n5870, n6201, and n6204.
- **Stability Analysis:** Dynamic Analysis showed AE2-025 does not contribute to any stability issues.
- **Reactive Power Assessment:** The AE2-025 meets the reactive power requirement at the high side of the facility transformer.

4 Revision to February 2020 System Impact Study Report

The AE2-025 System Impact Study is being revised to correct the scope and cost estimate for PJM Network Upgrade Number n6559 (Build a second Conastone – Peach Bottom 500 kV line). The upgrade was identified as a required System Reinforcement in the Light Load Analysis section of this report. Only the PECO portion of the cost estimate was included in the original report from February 2020. The correction includes estimates for both the PECO and BGE sections of the new line. See Section 15.5 for “System Reinforcements – Light Load” for the updated scope and cost for PJM Network Upgrade n6559.

5 General

The Interconnection Customer (IC), has proposed a Offshore Wind generating facility located in Ocean County, New Jersey. The installed facilities will have a total capability of 445.2 MW with 78.36 MW of this output being recognized by PJM as Capacity. The proposed in-service date for this project is December 1, 2025. This study does not imply a TO commitment to this in-service date.

Queue Number	AE2-025
Project Name	LARRABEE 230 KV II
Interconnection Customer	Atlantic Shores Offshore Wind Project 2, LLC
State	New Jersey
County	Ocean
Transmission Owner	JCPL
MFO	445.2
MWE	445.2
MWC	78.36

Queue Number	AE2-025
Fuel	Offshore Wind
Basecase Study Year	2022

6 Point of Interconnection

The interconnection of the project to the JCPL system will be accomplished by constructing a new Direct Connection to the Larrabee 230 kV substation. The IC will be responsible for acquiring all easements, properties, and permits that may be required to construct both the new interconnection switching station and the associated facilities. The project will also require Direct Connection upgrades at the Larrabee 230 kV substation.

Attachment 1 shows a one-line diagram of the proposed Direct Connection facilities for the AE2-025 generation project to connect to the FirstEnergy (“FE”) transmission system. **Attachment 2** provides the proposed location for the point of interconnection. IC will be responsible for constructing all the facilities on its side of the POI, including the Attachment Facilities which connect the generator to the FE transmission system’s Direct Connection facilities.

7 Cost Summary

The AE2-025 project will be responsible for the following costs:

Description	Total Cost
Attachment Facilities	\$352,500
Direct Connection Network Upgrade	\$0
Non Direct Connection Network Upgrades	\$4,034,900
System Upgrades (Summer Peak)	\$36,973,404 ¹
System Upgrades (Light Load)	\$0
Total Costs	\$41,360,804

The costs provided above exclude the Contribution in Aid of Construction (“CIAC”) Federal Income Tax Gross Up charge. If, at a future date, it is determined that the CIAC Federal Income Tax Gross Up charge is required, the Transmission Owner shall be reimbursed by the Interconnection Customer for such taxes.

The required Attachment Facilities and Direct and/or Non-Direct Connection work for the interconnection of the AE2-025 generation project to the FE Transmission System is detailed in the following sections. The associated one-line with the generation project Attachment Facilities and the Primary Direct and Non-Direct Connection facilities are shown in Attachment 1.

Note: PJM Open Access Transmission Tariff (OATT) section 217.3A outline cost allocation rules. The rules are further clarified in PJM Manual 14A Attachment B. The allocation of costs for a network upgrade will start with the first Queue project to cause the need for the upgrade. Later queue projects will receive cost allocation contingent on their contribution to the violation and are allocated to the queues that have not closed less than

¹ See “System Reinforcements – Summer Peak” Section 13 for details.

5 years following the execution of the first Interconnection Service Agreement which identifies the need for this upgrade.

8 Transmission Owner Scope of Work

The interconnection of the project to the JCPL system will be accomplished by constructing a new Direct Connection to the Larrabee 230 kV substation. The IC will be responsible for acquiring all easements, properties, and permits that may be required to construct both the new interconnection switching station and the associated facilities. The project will also require Direct Connection upgrades at the Larrabee 230 kV substation.

Attachment 1 shows a one-line diagram of the proposed Direct Connection facilities for the AE2-025 generation project to connect to the FirstEnergy (“FE”) transmission system. **Attachment 2** provides the proposed location for the point of interconnection. IC will be responsible for constructing all the facilities on its side of the POI, including the Attachment Facilities which connect the generator to the FE transmission system’s Direct Connection facilities.

8.1 Attachment Facilities

The total preliminary cost estimate for the Attachment work is given in the table below. These costs do not include CIAC Tax Gross-up.

Description	Total Cost
Install Attachment Facilities associated with the project AE2-025 at Larrabee 230 kV substation. Project is an offshore wind generator connecting at Larrabee 230 kV substation	\$ 281,600
Review nameplates, drawings, and relay settings at AE2-025 customer station	\$ 70,900
Total Attachment Facility Costs	\$352,500

8.2 Direct Connection Cost Estimate

The total preliminary cost estimate for the Direct Connection work is given in the table below. These costs do not include CIAC Tax Gross-up.

Description	Total Cost
	\$0
Total Direct Connection Facility Costs	\$0

8.3 Non-Direct Connection Cost Estimate

The total preliminary cost estimate for the Non-Direct Connection work is given in the table below. These costs do not include CIAC Tax Gross-up.

Description	Total Cost
Direct Connection for generation queue project AE2-025 at Larrabee 230 kV substation. Project is an offshore wind generator connecting at Larrabee 230 kV substation	\$ 1,519,200
Re-terminate the D2004 Larrabee-Smithburg No. 1 230 kV line to accommodate the updated Larrabee 230 kV substation providing a Direct Connection for generation queue project AE2-025 at D2004 Larrabee- Smithburg No. 1 230 kV line	\$ 1,914,900
Transfer existing conductor on the H2008 (Larrabee-Smithburg No. 2) 230 kV line to accommodate the updated Larrabee substation at H2008 Larrabee-Smithburg No. 2 230 kV line	\$ 244,500
Estimated SCADA work at Larrabee 230 kV substation to support breaker installations, relay installations, and updated relay settings. Estimated in-sub fiber run to provide communications and control from the AE2-025 control house to the nearest generation T-Line structure for fiber build.	\$ 59,700
Project Management, Environmental, and Forestry.	\$ 296,600
Total Non-Direct Connection Facility Costs	\$4,034,900

9 Schedule

Based on the scope of work for the Attachment Facilities and the Direct and/or Non-Direct Connection facilities, it is expected to take a minimum of **16 months** after the signing of an Interconnection Construction Service Agreement to complete the installation. This includes the requirement for the IC to make a preliminary payment that compensates FE for the first three months of the engineering design work that is related to the Attachment Facilities. Full initial deposit is required for the Non-Direct Connection work. This assumes that there will be no environmental issues with any of the new properties associated with this project, that there will be no delays in acquiring the necessary permits for implementing the defined Direct Connection and network upgrades, and that all transmission system outages will be allowed when requested.

The schedule for the required Network Impact Reinforcements will be more clearly identified in future study phases. The estimate elapsed time to complete each of the required reinforcements is identified in the “System Reinforcements” section of the report.

10 Transmission Owner Analysis

10.1 Power Flow Analysis

FE performed an analysis of its underlying transmission <100 kV system. The AE2-025 project did not contribute to any overloads on the FE transmission <100 kV system.

11 Interconnection Customer Requirements

11.1 System Protection

The IC must design its Customer Facilities in accordance with all applicable standards, including the standards in FE's "Requirements for Transmission Connected Facilities" document located at: <http://www.pjm.com/planning/design-engineering/to-tech-standards/private-firstenergy.aspx>.

Preliminary protection requirements will be provided as part of the Facilities Study. Detailed protection requirements will be provided once the project enters the construction phase.

11.2 Compliance Issues and Interconnection Customer Requirements

The proposed Customer Facilities must be designed in accordance with FE's "Requirements for Transmission Connected Facilities" document located at: <http://www.pjm.com/planning/design-engineering/to-tech-standards/private-firstenergy.aspx>. In particular, the IC is responsible for the following:

1. The purchase and installation of a fully rated 230 kV circuit breaker to protect the AE2-025 generator lead line. A single circuit breaker must be used to protect this line; if the project has several GSU transformers, the individual GSU transformer breakers cannot be used to protect this line.
2. The purchase and installation of the minimum required FE generation interconnection relaying and control facilities. This includes over/under voltage protection, over/under frequency protection, and zero sequence voltage protection relays.
3. The purchase and installation of supervisory control and data acquisition ("SCADA") equipment to provide information in a compatible format to the FE Transmission System Control Center.
4. Compliance with the FE and PJM generator power factor and voltage control requirements.
5. The execution of a back-up service agreement to serve the customer load supplied from the AE2-025 generation project metering point when the units are out-of-service. This assumes the intent of the IC is to net the generation with the load.

The IC will also be required to meet all PJM, ReliabilityFirst, and NERC reliability criteria and operating procedures for standards compliance. For example, the IC will need to properly locate and report the over and under voltage and over and under frequency system protection elements for its units as well as the submission of the generator model and protection data required to satisfy the PJM and ReliabilityFirst audits. Failure to comply with these requirements may result in a disconnection of service if the violation is found to compromise the reliability of the FE system.

11.3 Power Factor Requirements

The IC shall design its non-synchronous Customer Facility with the ability to maintain a power factor of at least 0.95 leading (absorbing VARs) to 0.95 lagging (supplying VARs) measured at the high side of the facility substation transformer(s) connected to the FE transmission system.

12 Revenue Metering and SCADA Requirements

12.1 PJM Requirements

The Interconnection Customer will be required to install equipment necessary to provide Revenue Metering (KWH, KVARH) and real time data (KW, KVAR) for IC's generating Resource. See PJM Manuals M-01 and M-14D, and PJM Tariff Section 8 of Attachment O.

12.2 FirstEnergy Requirements

The IC will be required to comply with all FE revenue metering requirements for generation interconnection customers which can be found in FE's "Requirements for Transmission Connected Facilities" document located at: <http://www.pjm.com/planning/design-engineering/to-tech-standards/private-firstenergy.aspx>.

13 Network Impacts – Summer Peak

The Queue Project AE2-025 was evaluated as a 445.2 MW (Capacity 78.4 MW) injection at the Larrabee 230 kV substation in the JCPL area. Project AE2-025 was evaluated for compliance with applicable reliability planning criteria (PJM, NERC, NERC Regional Reliability Councils, and Transmission Owners). Project AE2-025 was studied with a commercial probability of 100%. Potential network impacts were as follows:

Summer Peak Load Flow

13.1 Generation Deliverability

(Single or N-1 contingencies for the Capacity portion only of the interconnection)

None

13.2 Multiple Facility Contingency

(Double Circuit Tower Line, Fault with a Stuck Breaker, and Bus Fault contingencies for the full energy output)

Contingency			Affected Area	Facility Description	Bus			Power Flow	Loading %		Rating		MW Contribution	Flowgate Appendix
No.	Type	Name			From	To	Circuit		Initial	Final	Type	MVA		
1	LFFB	JC-P2-3-JCC-230-026A	JCPL - JCPL	28RED OAKA-28RAR RVR 230 kV line	2063 14	2063 05	1	AC	92.9	102.64	ER	869	84.22	
2	LFFB	JC-P2-3-JCC-230-026C	JCPL - JCPL	28RED OAKB-28RAR RVR 230 kV line	2063 15	2063 05	1	AC	97.56	107.24	ER	869	83.79	

13.3 Contribution to Previously Identified Overloads

(This project contributes to the following contingency overloads, i.e. "Network Impacts", identified for earlier generation or transmission interconnection projects in the PJM Queue)

Contingency			Affected Area	Facility Description	Bus			Power Flow	Loading %		Rating		MW Contribution	Flowgate Appendix
No.	Type	Name			From	To	Circuit		Initial	Final	Type	MVA		
3	LFFB	PS_P2-3_LNEPS_P1-2_2_LT	JCPL - PSEG	28RAR RVR-KILMER_I 230 kV line	2063 05	2183 31	1	AC	100.28	108.94	ER	813	82.64	1
4	LFFB	JC-P2-3-JCC-230-26B	JCPL - JCPL	28RED OAKA-28RAR RVR 230 kV line	2063 14	2063 05	1	AC	105.84	115.55	ER	869	83.89	2
5	LFFB	JC-P2-3-JCC-230-26D	JCPL - JCPL	28RED OAKB-28RAR RVR 230 kV line	2063 15	2063 05	1	AC	110.36	120.04	ER	869	83.47	3

13.4 Potential Congestion due to Local Energy Deliverability

PJM also studied the delivery of the energy portion of this interconnection request. Any problems identified below are likely to result in operational restrictions to the project under study. The developer can proceed with network upgrades to eliminate the operational restriction at their discretion by submitting a Merchant Transmission Interconnection request.

Note: Only the most severely overloaded conditions are listed below. There is no guarantee of full delivery of energy for this project by fixing only the conditions listed in this section. With a Transmission Interconnection Request, a subsequent analysis will be performed which shall study all overload conditions associated with the overloaded element(s) identified.

Contingency					Bus		Loading %			Rating				
No.	Type	Name	Affected Area	Facility Description	From	To	Circuit	Power Flow	Initial	Final	Type	MVA	MW Contribution	Flowgate Appendix
6	N-1	PS_P1-2_#2LINE_LT	JCPL - PSEG	28RAR RVR-KILMER_I 230 kV line	2063 05	2183 31	1	AC	100.28	108.94	ER	813	82.64	
7	N-1	JC-P1-2-JCC-230-028	JCPL - JCPL	28RED OAKB-28RAR RVR 230 kV line	2063 15	2063 05	1	AC	92.07	101.69	ER	869	83.27	
8	N-1	PS_P1-2_#2LINE	PSEG - PSEG	KILMER_I-LNELSN_I 230 kV line	2183 31	2183 33	1	AC	95.21	103.97	ER	805	82.64	
9	N-1	PS_P1-2_#1LINE	PSEG - PSEG	KILMER_W-LNELSN_W 230 kV line	2183 32	2183 34	1	AC	97.27	105.07	ER	679	62.16	

13.5 New System Reinforcements – Summer Peak

(Upgrades required to mitigate reliability criteria violations, i.e. Network Impacts, initially caused by the addition of this project generation)

Violation #	Overloaded Facility	Upgrade Description	Network Upgrade Number	Upgrade Cost	AE2-025 Allocation
1	28RED OAKA-28RAR RVR 230 kV line	AE2-025 is the driver of the overload of this facility for contingency 'JC-P2-3-JCC-230-026A' and also a contributor for a more heavily loaded contingency. See section 13.6 for the reinforcement and costs, if any.	-	-	-
2	28RED OAKB-28RAR RVR 230 kV line	AE2-025 is the driver of the overload of this facility for contingency 'JC-P2-3-JCC-230-026C' and also a contributor for a more heavily loaded contingency. See section 13.6 for the reinforcement and costs, if any.	-	-	-
Total Cost (\$)				\$0	\$0

13.6 Contribution to Previously Identified System Reinforcements – Summer Peak

Violation #	Overloaded Facility	Upgrade Description	Network Upgrade Number	Upgrade Cost	AE2-025 Allocation
4	28RAR RVR-KILMER_I	PJM Baseline: Project Id: b3042 Project Description: Replace substation conductor at Raritan River 203 kV substation on the Kilmer line terminal. Project Type: FAC New Ratings: Rate A: 709 MVA Rate B: 869 MVA Rate C: 985 MVA Estimated in service: 6/1/2023 Note 1: Although Queue Project AE2-025 may not have cost responsibility for this upgrade, Queue Project AE2-025 may need this upgrade in-service to be deliverable to the PJM system. If Queue Project AE2-025 comes into service prior to completion of the upgrade, Queue Project AE2-025 will need an interim study.	b3042	-	\$0

		JCPL Project Id: n6201 Project Description: Reconductor the Kilmer (I1023) - Raritan River 230 kV transmission line (~5.92 miles). Upgrade terminals as required Project Type: FAC New Ratings: Rate A: 1156 MVA Rate B: 1334 MVA Rate C: 1334 MVA Cost: \$27,302,400 Time estimate: 42 months. AE2-025 is the driver for this overload and reinforcement of the facility after baseline b3042 is in service.	n6201	\$27,302,400	\$27,302,400												
5	28RED OAKA-28RAR RVR 230 kV line	JCPL Project Id: n5870 Project Description: Reconductor the Red Oak Tap-Raritan River segment of the T1034 line, approximately 2.6 miles. Project Type: FAC New Ratings: Rate A: 1156 MVA Rate B: 1334 MVA Rate C: 1334 MVA Cost: \$12,528,000 Time estimate: 30 months. Cost allocation table: <table><tr><th>Queue Project</th><th>MW Impact</th><th>Cost (%)</th><th>Cost(\$)</th></tr><tr><td>AE2-024</td><td>50.42</td><td>37.54%</td><td>\$4,703,083</td></tr><tr><td>AE2-025</td><td>83.89</td><td>62.46%</td><td>\$7,824,917</td></tr></table>	Queue Project	MW Impact	Cost (%)	Cost(\$)	AE2-024	50.42	37.54%	\$4,703,083	AE2-025	83.89	62.46%	\$7,824,917	n5870	\$12,528,000	\$7,824,917
Queue Project	MW Impact	Cost (%)	Cost(\$)														
AE2-024	50.42	37.54%	\$4,703,083														
AE2-025	83.89	62.46%	\$7,824,917														
6	28RED OAKB-28RAR RVR 230 kV line	JCPL Project Id: n6204 Project Description: Red Oak B-Raritan River 230 kV Line: Reconductor existing sections of 1590 ACSR 45/7 (Segment 3, Position 2; Segment 7, Position 1) with 1590 ACSS 45/7. Project Type: FAC New Ratings: Rate A: 1156 MVA Rate B: 1334 MVA Rate C: 1334 MVA Cost: \$3,900,000 Time estimate: 30 months. Cost allocation table:	n6204	\$3,900,000	\$1,846,087												

		<table><tr><th>Queue Project</th><th>MW Impact</th><th>Cost (%)</th><th>Cost (\$)</th></tr><tr><td>AE2-024</td><td>92.80</td><td>52.7%</td><td>\$2,053,913</td></tr><tr><td>AE2-025</td><td>83.41</td><td>47.3%</td><td>\$1,846,087</td></tr></table>	Queue Project	MW Impact	Cost (%)	Cost (\$)	AE2-024	92.80	52.7%	\$2,053,913	AE2-025	83.41	47.3%	\$1,846,087			
Queue Project	MW Impact	Cost (%)	Cost (\$)														
AE2-024	92.80	52.7%	\$2,053,913														
AE2-025	83.41	47.3%	\$1,846,087														
Total Cost (\$)					\$43,730,400	\$36,973,404											

13.7 Flow Gate Details

The following indices contain additional information about each flowgate presented in the body of the report. For each appendix, a description of the flowgate and its contingency was included for convenience. However, the intent of the appendix section is to provide more information on which projects/generators have contributions to the flowgate in question. Although this information is not used "as is" for cost allocation purposes, it can be used to gage other generators impact. It should be noted the generator contributions presented in the appendices sections are full contributions, whereas in the body of the report, those contributions take into consideration the commercial probability of each project.

13.8 Index 1

(JCPL - PSEG) The 28RAR RVR-KILMER_I 230 kV line (from bus 206305 to bus 218331 ckt 1) loads from 100.28% to 108.94% (AC power flow) of its emergency rating (813 MVA) for the line fault with failed breaker contingency outage of 'PS_P2-3_LNEPS_P1-2_2_LT'. This project contributes approximately 82.64 MW to the thermal violation.

```

CONTINGENCY 'PS_P2-3_LNEPS_P1-2_2_LT'          /* LAKE NELSON LINE #2 STUCK
BREAKER
DISCONNECT BUS 218334                          /* LAKE NELSON LOSS OF BUS W SIDE
DISCONNECT BUS 218522                          /* REMOVE MIDDLESEX
DISCONNECT BUS 218332                          /* REMOVE KILMER
CLOSE LINE FROM BUS 218384 TO BUS 218387 CKT Z  /* KILMER
CLOSE LINE FROM BUS 218385 TO BUS 218386 CKT Z  /* KILMER
MOVE 8 MW LOAD FROM BUS 218384 TO BUS 218383    /* INTERSTATION TIE
TRANSFER LOAD FROM KILMER TO GREENBK T1
MOVE 8 MW LOAD FROM BUS 218387 TO BUS 218399    /* INTERSTATION TIE
TRANSFER LOAD FROM KILMER TO NEWDOVR T2
MOVE 8 MW LOAD FROM BUS 218385 TO BUS 218393    /* INTERSTATION TIE
TRANSFER LOAD FROM KILMER TO MDWRD T2
MOVE 8 MW LOAD FROM BUS 218386 TO BUS 218347    /* INTERSTATION TIE
TRANSFER LOAD FROM KILMER TO BNNTLANE T2
CLOSE LINE FROM BUS 218388 TO BUS 218389 CKT Z  /* LAKE NELSON
MOVE 8 MW LOAD FROM BUS 218389 TO BUS 218384    /* INTERSTATION TIE
TRANSFER LOAD FROM LNELSN TO KILMER T2
MOVE 8 MW LOAD FROM BUS 218388 TO BUS 218385    /* INTERSTATION TIE
TRANSFER LOAD FROM LNELSN TO KILMER T4
END

```

<i>Bus Number</i>	<i>Bus Name</i>	<i>Full Contribution</i>
206362	28RDOAKCT1	13.42
206363	28RDOAKCT2	13.39
206364	28RDOAKCT3	13.39
206365	28RDOAKST1	18.81
206350	28RRCT1&2	6.35
206351	28RRCT3&4	5.94
206329	28S RIV G3	7.35
206403	28W4-009	58.14
939121	AE1-142 C O1	1.37

939122	AE1-142 E O1	1.97
939981	AE1-238 C	61.08
939982	AE1-238 E	160.44
940161	AE2-000 C O1	35.34
940162	AE2-000 E O1	90.42
940401	AE2-024 C O1	28.82
940402	AE2-024 E O1	134.91
940411	AE2-025 C	14.55
940412	AE2-025 E	68.1
940691	AE2-056 C	0.37
940692	AE2-056 E	0.33
940921	AE2-081 C	0.4
940922	AE2-081 E	0.41
LTF	BLUEG	3.31
LTF	CALDERWOOD	0.35
LTF	CANNELTON	0.2
LTF	CARR	1.12
LTF	CATAWBA	0.22
LTF	CHEOAH	0.32
LTF	COFFEEN	0.35
LTF	COTTONWOOD	1.35
LTF	DEARBORN /* 35% REVERSE 4540149	< 0.01
LTF	DUCKCREEK	0.76
LTF	EDWARDS	0.35
LTF	FARMERCITY	0.23
LTF	G-007A	124.14
LTF	GIBSON	0.14

<i>LTF</i>	<i>HAMLET</i>	<i>0.35</i>
<i>LTF</i>	<i>NEWTON</i>	<i>0.91</i>
<i>LTF</i>	<i>O-066</i>	<i>33.52</i>
<i>LTF</i>	<i>PRAIRIE</i>	<i>1.69</i>
<i>LTF</i>	<i>RENSSELAER</i>	<i>0.89</i>
<i>LTF</i>	<i>ROWAN</i> /* 35% REVERSE 4479078	<i>< 0.01</i>
<i>LTF</i>	<i>SMITHLAND</i>	<i>0.13</i>
<i>LTF</i>	<i>TILTON</i>	<i>0.42</i>
<i>LTF</i>	<i>TRIMBLE</i>	<i>0.37</i>
<i>LTF</i>	<i>TVA</i>	<i>1.13</i>
<i>LTF</i>	<i>UNIONPOWER</i>	<i>0.5</i>
<i>901032</i>	<i>W1-024E OPI</i>	<i>0.46</i>
<i>901112</i>	<i>W1-032 E OPI</i>	<i>0.28</i>
<i>901912</i>	<i>W1-112E OPI</i>	<i>0.84</i>
<i>901422</i>	<i>W1-113 E</i>	<i>0.85</i>
<i>901982</i>	<i>W1-119 E</i>	<i>1.47</i>
<i>901991</i>	<i>W1-120C</i>	<i>1.</i>
<i>901992</i>	<i>W1-120E</i>	<i>1.63</i>
<i>902032</i>	<i>W1-124E</i>	<i>3.46</i>
<i>902082</i>	<i>W1-129E</i>	<i>0.39</i>
<i>902322</i>	<i>W2-019 E</i>	<i>0.47</i>
<i>902912</i>	<i>W2-078 E</i>	<i>1.05</i>
<i>902952</i>	<i>W2-082 E OPI</i>	<i>1.34</i>
<i>903982</i>	<i>W3-079 E</i>	<i>0.86</i>
<i>905252</i>	<i>W4-025 E</i>	<i>0.56</i>
<i>905502</i>	<i>W4-060 E OPI</i>	<i>2.39</i>
<i>907082</i>	<i>X1-037 E</i>	<i>2.45</i>

907272	X1-085 E	0.41
912102	X4-015 E	0.39
912182	X4-031 E	0.5
914092	Y2-051 E	0.6
917612	Z2-102 E	1.07
918452	AA1-060 E	2.57
919662	AA2-048 E	1.38
920732	AA2-184 E	5.62
930891	AB1-138 C	0.61
930892	AB1-138 E	1.02
931122	AB1-163 E	0.84
924852	AB2-139 E	0.08
927131	AC1-207 C	1.
927132	AC1-207 E	1.68

13.9 Index 3

(JCPL - JCPL) The 28RED OAKA-28RAR RVR 230 kV line (from bus 206314 to bus 206305 ckt 1) loads from 105.84% to 115.55% (AC power flow) of its emergency rating (869 MVA) for the line fault with failed breaker contingency outage of 'JC-P2-3-JCC-230-26B'. This project contributes approximately 83.89 MW to the thermal violation.

CONTINGENCY 'JC-P2-3-JCC-230-26B'
BREAKER

/* RARITAN RIVER B144 STUCK

DISCONNECT BUS 206315

/* RED OAK 230 KV B TAP

DISCONNECT BUS 206362

/* RED OAK 230 KV #1

DISCONNECT BUS 206365

/* RED OAK 230 KV #4

DISCONNECT BRANCH FROM BUS 206305 TO BUS 206303 CKT 13

/* RARITAN RIVER #13

230/115 KV TRANSFORMER

DISCONNECT BUS 206403

/* WOODBRIDGE GENERATOR

END

<i>Bus Number</i>	<i>Bus Name</i>	<i>Full Contribution</i>
207145	28FRN_Y2-051	0.06

207204	28HOL_W1-112	0.07
207144	28HOW_X1-037	0.21
206201	28JERSVGEN	0.06
206367	28LKWD CT2	4.45
206358	28PARLN1&2	4.13
206359	28PARLN3&4	5.09
206363	28RDOAKCT2	17.26
206364	28RDOAKCT3	17.26
206327	28S RIV G1	6.74
206328	28S RIV G2	6.74
206329	28S RIV G3	9.31
207206	28TIN_W1-124	0.35
939121	AE1-142 C O1	1.39
939122	AE1-142 E O1	2.
939981	AE1-238 C	67.4
939982	AE1-238 E	177.03
940161	AE2-000 C O1	35.92
940162	AE2-000 E O1	91.92
940401	AE2-024 C O1	29.25
940402	AE2-024 E O1	136.95
940411	AE2-025 C	14.77
940412	AE2-025 E	69.13
940691	AE2-056 C	0.33
940692	AE2-056 E	0.29
940921	AE2-081 C	0.35
940922	AE2-081 E	0.36
LTF	BLUEG	1.33

<i>LTF</i>	<i>CALDERWOOD</i>	<i>0.13</i>
<i>LTF</i>	<i>CANNELTON</i>	<i>0.08</i>
<i>LTF</i>	<i>CARR</i>	<i>1.03</i>
<i>LTF</i>	<i>CATAWBA</i>	<i>0.08</i>
<i>LTF</i>	<i>CHEOAH</i>	<i>0.12</i>
<i>LTF</i>	<i>COFFEEN</i>	<i>0.14</i>
<i>LTF</i>	<i>COTTONWOOD</i>	<i>0.53</i>
<i>LTF</i>	<i>DUCKCREEK</i>	<i>0.31</i>
<i>LTF</i>	<i>EDWARDS</i>	<i>0.14</i>
<i>LTF</i>	<i>FARMERCITY</i>	<i>0.09</i>
<i>LTF</i>	<i>G-007</i>	<i>30.08</i>
<i>LTF</i>	<i>GIBSON</i>	<i>0.06</i>
<i>LTF</i>	<i>HAMLET</i>	<i>0.12</i>
<i>LTF</i>	<i>NEWTON</i>	<i>0.37</i>
<i>LTF</i>	<i>O-066</i>	<i>33.85</i>
<i>LTF</i>	<i>PRAIRIE</i>	<i>0.67</i>
<i>LTF</i>	<i>RENSSELAER</i>	<i>0.81</i>
<i>LTF</i>	<i>SMITHLAND</i>	<i>0.05</i>
<i>LTF</i>	<i>TILTON</i>	<i>0.17</i>
<i>LTF</i>	<i>TRIMBLE</i>	<i>0.15</i>
<i>LTF</i>	<i>TVA</i>	<i>0.44</i>
<i>LTF</i>	<i>UNIONPOWER</i>	<i>0.19</i>
<i>901032</i>	<i>W1-024E OPI</i>	<i>0.24</i>
<i>901112</i>	<i>W1-032 E OPI</i>	<i>0.2</i>
<i>901912</i>	<i>W1-112E OPI</i>	<i>0.76</i>
<i>901982</i>	<i>W1-119 E</i>	<i>1.42</i>
<i>901991</i>	<i>W1-120C</i>	<i>0.97</i>

<i>901992</i>	<i>W1-120E</i>	<i>1.57</i>
<i>902032</i>	<i>W1-124E</i>	<i>3.66</i>
<i>902082</i>	<i>W1-129E</i>	<i>0.38</i>
<i>902322</i>	<i>W2-019 E</i>	<i>0.45</i>
<i>902952</i>	<i>W2-082 E OP1</i>	<i>1.29</i>
<i>903981</i>	<i>W3-079 C</i>	<i>0.08</i>
<i>903982</i>	<i>W3-079 E</i>	<i>0.88</i>
<i>905252</i>	<i>W4-025 E</i>	<i>0.54</i>
<i>907082</i>	<i>X1-037 E</i>	<i>2.16</i>
<i>912102</i>	<i>X4-015 E</i>	<i>0.38</i>
<i>912182</i>	<i>X4-031 E</i>	<i>0.51</i>
<i>914092</i>	<i>Y2-051 E</i>	<i>0.61</i>
<i>917612</i>	<i>Z2-102 E</i>	<i>1.04</i>
<i>918452</i>	<i>AA1-060 E</i>	<i>2.14</i>
<i>919662</i>	<i>AA2-048 E</i>	<i>1.4</i>
<i>920732</i>	<i>AA2-184 E</i>	<i>5.92</i>
<i>930891</i>	<i>AB1-138 C</i>	<i>0.62</i>
<i>930892</i>	<i>AB1-138 E</i>	<i>1.03</i>
<i>931122</i>	<i>AB1-163 E</i>	<i>0.84</i>

13.10 Index 4

(JCPL - JCPL) The 28RED OAKB-28RAR RVR 230 kV line (from bus 206315 to bus 206305 ckt 1) loads from 110.36% to 120.04% (AC power flow) of its emergency rating (869 MVA) for the line fault with failed breaker contingency outage of 'JC-P2-3-JCC-230-26D'. This project contributes approximately 83.47 MW to the thermal violation.

```
CONTINGENCY 'JC-P2-3-JCC-230-26D'          /* RARITAN RIVER B146 STUCK
BREAKER
DISCONNECT BUS 206314                      /* RED OAK 230 KV A TAP
DISCONNECT BUS 206363                      /* RED OAK 230 KV #2
DISCONNECT BUS 206364                      /* RED OAK 230 KV #3
DISCONNECT BRANCH FROM BUS 206305 TO BUS 206303 CKT 13 /* RARITAN RIVER #13
230/115 KV TRANSFORMER
DISCONNECT BUS 206403                      /* WOODBRIDGE GENERATOR
END
```

<i>Bus Number</i>	<i>Bus Name</i>	<i>Full Contribution</i>
207145	28FRN_Y2-051	0.06
207204	28HOL_W1-112	0.07
207144	28HOW_X1-037	0.21
206201	28JERSVGEN	0.06
206367	28LKWD CT2	4.43
206358	28PARLN1&2	4.18
206359	28PARLN3&4	5.16
206362	28RDOAKCT1	17.3
206365	28RDOAKST1	24.25
206327	28S RIV G1	6.6
206328	28S RIV G2	6.6
206329	28S RIV G3	9.12
207206	28TIN_W1-124	0.35
939121	AE1-142 C O1	1.38
939122	AE1-142 E O1	1.99
939981	AE1-238 C	67.08
939982	AE1-238 E	176.19

<i>940161</i>	<i>AE2-000 C OI</i>	<i>35.73</i>
<i>940162</i>	<i>AE2-000 E OI</i>	<i>91.43</i>
<i>940401</i>	<i>AE2-024 C OI</i>	<i>29.1</i>
<i>940402</i>	<i>AE2-024 E OI</i>	<i>136.25</i>
<i>940411</i>	<i>AE2-025 C</i>	<i>14.69</i>
<i>940412</i>	<i>AE2-025 E</i>	<i>68.78</i>
<i>940691</i>	<i>AE2-056 C</i>	<i>0.33</i>
<i>940692</i>	<i>AE2-056 E</i>	<i>0.28</i>
<i>940921</i>	<i>AE2-081 C</i>	<i>0.35</i>
<i>940922</i>	<i>AE2-081 E</i>	<i>0.36</i>
<i>LTF</i>	<i>BLUEG</i>	<i>1.45</i>
<i>LTF</i>	<i>CALDERWOOD</i>	<i>0.15</i>
<i>LTF</i>	<i>CANNELTON</i>	<i>0.09</i>
<i>LTF</i>	<i>CARR</i>	<i>1.03</i>
<i>LTF</i>	<i>CATAWBA</i>	<i>0.09</i>
<i>LTF</i>	<i>CHEOAH</i>	<i>0.13</i>
<i>LTF</i>	<i>COFFEEN</i>	<i>0.15</i>
<i>LTF</i>	<i>COTTONWOOD</i>	<i>0.58</i>
<i>LTF</i>	<i>DUCKCREEK</i>	<i>0.33</i>
<i>LTF</i>	<i>EDWARDS</i>	<i>0.15</i>
<i>LTF</i>	<i>FARMERCITY</i>	<i>0.1</i>
<i>LTF</i>	<i>G-007</i>	<i>29.99</i>
<i>LTF</i>	<i>GIBSON</i>	<i>0.06</i>
<i>LTF</i>	<i>HAMLET</i>	<i>0.14</i>
<i>LTF</i>	<i>NEWTON</i>	<i>0.4</i>
<i>LTF</i>	<i>O-066</i>	<i>33.89</i>
<i>LTF</i>	<i>PRAIRIE</i>	<i>0.73</i>

<i>LTF</i>	<i>RENSSELAER</i>	<i>0.82</i>
<i>LTF</i>	<i>SMITHLAND</i>	<i>0.06</i>
<i>LTF</i>	<i>TILTON</i>	<i>0.18</i>
<i>LTF</i>	<i>TRIMBLE</i>	<i>0.16</i>
<i>LTF</i>	<i>TVA</i>	<i>0.48</i>
<i>LTF</i>	<i>UNIONPOWER</i>	<i>0.21</i>
<i>901032</i>	<i>W1-024E OPI</i>	<i>0.24</i>
<i>901112</i>	<i>W1-032 E OPI</i>	<i>0.2</i>
<i>901912</i>	<i>W1-112E OPI</i>	<i>0.76</i>
<i>901982</i>	<i>W1-119 E</i>	<i>1.41</i>
<i>901991</i>	<i>W1-120C</i>	<i>0.96</i>
<i>901992</i>	<i>W1-120E</i>	<i>1.57</i>
<i>902032</i>	<i>W1-124E</i>	<i>3.65</i>
<i>902082</i>	<i>W1-129E</i>	<i>0.38</i>
<i>902322</i>	<i>W2-019 E</i>	<i>0.44</i>
<i>902952</i>	<i>W2-082 E OPI</i>	<i>1.28</i>
<i>903981</i>	<i>W3-079 C</i>	<i>0.08</i>
<i>903982</i>	<i>W3-079 E</i>	<i>0.87</i>
<i>905252</i>	<i>W4-025 E</i>	<i>0.53</i>
<i>907082</i>	<i>X1-037 E</i>	<i>2.16</i>
<i>912102</i>	<i>X4-015 E</i>	<i>0.38</i>
<i>912182</i>	<i>X4-031 E</i>	<i>0.51</i>
<i>914092</i>	<i>Y2-051 E</i>	<i>0.61</i>
<i>917612</i>	<i>Z2-102 E</i>	<i>1.03</i>
<i>918452</i>	<i>AA1-060 E</i>	<i>2.12</i>
<i>919662</i>	<i>AA2-048 E</i>	<i>1.39</i>
<i>920732</i>	<i>AA2-184 E</i>	<i>5.9</i>

<i>930891</i>	<i>ABI-138 C</i>	<i>0.62</i>
<i>930892</i>	<i>ABI-138 E</i>	<i>1.03</i>
<i>931122</i>	<i>ABI-163 E</i>	<i>0.83</i>

Affected Systems

13.11 Affected Systems

13.11.1 NYISO

None.

13.12 Contingency Descriptions

Contingency Name	Description
JC-P1-2-JCC-230-028	CONTINGENCY 'JC-P1-2-JCC-230-028' /* RARITAN RIVER - RED OAK - SOUTH RIVER JCT(T1034) 230 KV DISCONNECT BRANCH FROM BUS 206305 TO BUS 206314 CKT 1 /* 28RAR RVR 230.00 28RED OAKA 230.00 DISCONNECT BRANCH FROM BUS 206314 TO BUS 206411 CKT 1 /* 28RED OAKA 230.00 28R11RINGA 230.00 . DISCONNECT BUS 206314 /* 28RED OAKA 230.00 DISCONNECT BUS 206363 /* 28RDOAKCT2 18.00 DISCONNECT BUS 206364 /* 28RDOAKCT3 18.00 END
JC-P2-3-JCC-230-026A	CONTINGENCY 'JC-P2-3-JCC-230-026A' /* RARITAN RIVER B143 STUCK BREAKER DISCONNECT BUS 206315 /* RED OAK 230 KV B TAP DISCONNECT BUS 206362 /* RED OAK 230 KV #1 DISCONNECT BUS 206365 /* RED OAK 230 KV #4 DISCONNECT BRANCH FROM BUS 206305 TO BUS 206303 CKT 17 /* RARITAN RIVER #17 230/115 KV TRANSFORMER DISCONNECT BUS 206350 /* SAYREVILLE GENERATOR DISCONNECT BUS 206351 /* SAYREVILLE GENERATOR END
JC-P2-3-JCC-230-026C	CONTINGENCY 'JC-P2-3-JCC-230-026C' /* RARITAN RIVER B145 STUCK BREAKER DISCONNECT BUS 206314 /* RED OAK 230 KV A TAP DISCONNECT BUS 206363 /* RED OAK 230 KV #2 DISCONNECT BUS 206364 /* RED OAK 230 KV #3 DISCONNECT BRANCH FROM BUS 206305 TO BUS 206303 CKT 17 /* RARITAN RIVER #17 230/115 KV TRANSFORMER DISCONNECT BUS 206350 /* SAYREVILLE GENERATOR DISCONNECT BUS 206351 /* SAYREVILLE GENERATOR END
JC-P2-3-JCC-230-26B	CONTINGENCY 'JC-P2-3-JCC-230-26B' /* RARITAN RIVER B144 STUCK BREAKER DISCONNECT BUS 206315 /* RED OAK 230 KV B TAP DISCONNECT BUS 206362 /* RED OAK 230 KV #1 DISCONNECT BUS 206365 /* RED OAK 230 KV #4 DISCONNECT BRANCH FROM BUS 206305 TO BUS 206303 CKT 13 /* RARITAN RIVER #13

PS_P1-2_#2LINE_LT	CONTINGENCY 'PS_P1-2_#2LINE_LT'	/* LAKE NELSON TO RARITAN RIVER
	DISCONNECT BUS 218332	/* REMOVE KILMER
	CLOSE LINE FROM BUS 218384 TO BUS 218387 CKT Z	/* KILMER
	CLOSE LINE FROM BUS 218385 TO BUS 218386 CKT Z	/* KILMER
	MOVE 8 MW LOAD FROM BUS 218384 TO BUS 218383 LOAD FROM KILMER TO GREENBK T1	/* INTERSTATION TIE TRANSFER
	MOVE 8 MW LOAD FROM BUS 218387 TO BUS 218399 LOAD FROM KILMER TO NEWDOVR T2	/* INTERSTATION TIE TRANSFER
	MOVE 8 MW LOAD FROM BUS 218385 TO BUS 218393 LOAD FROM KILMER TO MDWRD T2	/* INTERSTATION TIE TRANSFER
	MOVE 8 MW LOAD FROM BUS 218386 TO BUS 218347 LOAD FROM KILMER TO BNNTLANE T2	/* INTERSTATION TIE TRANSFER
PS_P2-3_LNEPS_P1-2_2_LT	END	
	CONTINGENCY 'PS_P2-3_LNEPS_P1-2_2_LT'	/* LAKE NELSON LINE #2 STUCK BREAKER
	DISCONNECT BUS 218334	/* LAKE NELSON LOSS OF BUS W SIDE
	DISCONNECT BUS 218522	/* REMOVE MIDDLESEX
	DISCONNECT BUS 218332	/* REMOVE KILMER
	CLOSE LINE FROM BUS 218384 TO BUS 218387 CKT Z	/* KILMER
	CLOSE LINE FROM BUS 218385 TO BUS 218386 CKT Z	/* KILMER
	MOVE 8 MW LOAD FROM BUS 218384 TO BUS 218383 LOAD FROM KILMER TO GREENBK T1	/* INTERSTATION TIE TRANSFER
	MOVE 8 MW LOAD FROM BUS 218387 TO BUS 218399 LOAD FROM KILMER TO NEWDOVR T2	/* INTERSTATION TIE TRANSFER
	MOVE 8 MW LOAD FROM BUS 218385 TO BUS 218393 LOAD FROM KILMER TO MDWRD T2	/* INTERSTATION TIE TRANSFER
	MOVE 8 MW LOAD FROM BUS 218386 TO BUS 218347 LOAD FROM KILMER TO BNNTLANE T2	/* INTERSTATION TIE TRANSFER
	CLOSE LINE FROM BUS 218388 TO BUS 218389 CKT Z	/* LAKE NELSON
	MOVE 8 MW LOAD FROM BUS 218389 TO BUS 218384 LOAD FROM LNELSN TO KILMER T2	/* INTERSTATION TIE TRANSFER
	MOVE 8 MW LOAD FROM BUS 218388 TO BUS 218385 LOAD FROM LNELSN TO KILMER T4	/* INTERSTATION TIE TRANSFER
	END	

Short Circuit

14 Short Circuit

The following Breakers are overduty:

None

Stability

15 Stability Analysis and Reactive Power Assessment

Generator Interconnection Request **AE2-025** is for a 445.2 MW Maximum Facility Output (MFO) offshore wind generating facility consisting of 46 Siemens Gamesa SG-10.0-193 15 MVA wind turbines connecting to the JCPL transmission system. The **AE2-025** offshore wind generating facility will be located off the coast of Ocean County, New Jersey.

The **AE2-025** generation project will directly interconnect into the existing Larrabee 230 kV substation in the Jersey Central Power & Light Company (JCPL) transmission system. The Point of Interconnection (POI) will be at a JCPL installed deadend structure at the Larrabee substation yard.

This report describes a dynamic simulation analysis of AE2-025 as part of the overall system impact study. The load flow scenario for the analysis was based on the RTEP 2022 peak load case, modified to include applicable queue projects. AE2-025 has been dispatched online at maximum power output, with approximately 1.0325 pu/1.041 pu voltage at the generator terminals.

AE2-025 was tested for compliance with NERC, PJM, Transmission Owner, and other applicable criteria. 149 contingencies were studied, each with a 20 second simulation time period (with 1.0 second initial run prior to any events). Studied faults included:

- a) Steady state operation (Category P0);
- b) Three phase faults with normal clearing time on the intact network (Category P1);
- c) Single phase to ground faults with delayed clearing due to a stuck breaker (Category P4);
- d) Single phase faults placed at 80% of the line with delayed (Zone 2) clearing at line end remote from the fault due to primary communications/relay failure (Category P5);
- e) Single phase to ground faults with normal clearing for common structure (Category P7).

For all 149 fault contingencies tested on the 2022 peak load case:

- a) AE2-025 was able to ride through the faults (except for faults where protective action trips a generator(s)).
- b) Post-contingency oscillations were positively damped with a damping margin of at least 3%.
- c) Following fault clearing, all bus voltages recover to a minimum of 0.7 per unit after 2.5 seconds (except where protective action isolates that bus).
- d) No transmission element trips, other than those either directly connected or designed to trip as a consequence of that fault.

AE2-025 was assessed for compliance with reactive power capability requirements using the supplied capability curves. Please note this is a new facility.

- Generation shall have the ability to maintain a power factor of at least 0.95 leading to 0.95 lagging at the high voltage side of facility transformer

The generating facility AE2-025 **meets** the reactive power requirement at the high side of facility transformers.

Light Load

16 Network Impacts - Light Load Analysis

The Queue Project AE2-024 was evaluated as a 882.0 MW (Capacity 155.23 MW) injection at Larrabee 230 kV Substation in the JCPL area. Project AE2-024 was evaluated for compliance with applicable reliability planning criteria (PJM, NERC, NERC Regional Reliability Councils, and Transmission Owners). Project AE2-024 was studied with a commercial probability of 100%. Potential network impacts were as follows:

16.1 Generation Deliverability

(Single or N-1 contingencies for the Capacity portion only of the interconnection)

Contingency					Bus		Loading %			Rating		MW Contribution	Flowgate Appendix	
No.	Type	Name	Affected Area	Facility Description	From	To	Circuit	Power Flow	Initial	Final	Type			MVA
1	N-1	JC-P1-2-JCC-230-027	JCPL - JCPL	28RED OAKA-28RAR RVR 230 kV line	2063 14	2063 05	1	AC	93.78	101.51	ER	869	66.43	1
2	N-1	JC-P1-2-JCC-230-028	JCPL - JCPL	28RED OAKB-28RAR RVR 230 kV line	2063 15	2063 05	1	AC	98.95	106.66	ER	869	65.98	2

16.2 Multiple Facility Contingency

(Double Circuit Tower Line, Fault with a Stuck Breaker, and Bus Fault contingencies for the full energy output)

Contingency				Bus				Loading %		Rating		MW Contribution	Flowgate Appendix	
No.	Type	Name	Affected Area	Facility Description	From	To	Circuit	Power Flow	Initial	Final	Type			MVA
3	LFFB	JC-P2-3-JCC-230-26D	JCPL - JCPL	28RED OAKB-28RAR RVR 230 kV line	2063 15	2063 05	1	AC	96.59	104.15	ER	869	66.5	
4	LFFB	JC-P2-3-JCC-230-026C	JCPL - JCPL	28RED OAKB-28RAR RVR 230 kV line	2063 15	2063 05	1	AC	96.59	104.15	ER	869	66.5	

16.3 Contribution to Previously Identified Overloads

(This project contributes to the following contingency overloads, i.e. "Network Impacts", identified for earlier generation or transmission interconnection projects in the PJM Queue)

None

16.4 Potential Congestion due to Local Energy Deliverability

PJM also studied the delivery of the energy portion of this interconnection request. Any problems identified below are likely to result in operational restrictions to the project under study. The developer can proceed with network upgrades to eliminate the operational restriction at their discretion by submitting a Merchant Transmission Interconnection request.

Note: Only the most severely overloaded conditions are listed below. There is no guarantee of full delivery of energy for this project by fixing only the conditions listed in this section. With a Transmission Interconnection Request, a subsequent analysis will be performed which shall study all overload conditions associated with the overloaded element(s) identified.

None

16.5 System Reinforcements – Light Load

Violation #	Overloaded Facility	Upgrade Description	Network Upgrade Number	Upgrade Cost	AE2-025 Allocation
1	28RED OAKA-28RAR RVR 230 kV line	This facility is more overloaded in summer peak. See the summer peak analysis for the reinforcement and cost allocation.	-	-	-
2, 3, 4	28RED OAKB-28RAR RVR 230 kV line	This facility is more overloaded in summer peak. See the summer peak analysis for the reinforcement and cost allocation.	-	-	-
Total Cost (\$)				-	-

16.6 Flow Gate Details

The following appendices contain additional information about each flowgate presented in the body of the report. For each appendix, a description of the flowgate and its contingency was included for convenience. However, the intent of the appendix section is to provide more information on which projects/generators have contributions to the flowgate in question. Although this information is not used "as is" for cost allocation purposes, it can be used to gauge other generators impact. It should be noted the generator contributions presented in the appendices sections are full contributions, whereas in the body of the report, those contributions take into consideration the commercial probability of each project.

16.7 Index 1

(JCPL - JCPL) The 28RED OAKA-28RAR RVR 230 kV line (from bus 206314 to bus 206305 ckt 1) loads from 93.78% to 101.51% (AC power flow) of its emergency rating (869 MVA) for the single line contingency outage of 'JC-P1-2-JCC-230-027'. This project contributes approximately 66.43 MW to the thermal violation.

CONTINGENCY 'JC-P1-2-JCC-230-027'

/* RARITAN RIVER - RED OAK - R11 (G1047) 230 KV

LINE

DISCONNECT BRANCH FROM BUS 206305 TO BUS 206315 CKT 1

DISCONNECT BRANCH FROM BUS 206315 TO BUS 206410 CKT 1

DISCONNECT BUS 206315

DISCONNECT BUS 206362

DISCONNECT BUS 206365

END

<i>Bus Number</i>	<i>Bus Name</i>	<i>Full Contribution</i>
937512	AD2-210 E	0.13
938781	AE1-104 C OI	5.07
938782	AE1-104 E OI	12.97
939981	AE1-238 C	53.75
939982	AE1-238 E	141.19
940161	AE2-000 C OI	28.43
940162	AE2-000 E OI	72.73
940361	AE2-020 C OI	5.31
940362	AE2-020 E OI	24.87
940371	AE2-021 C OI	5.31
940372	AE2-021 E OI	24.87
940381	AE2-022 C OI	2.63
940382	AE2-022 E OI	12.34
940401	AE2-024 C OI	23.16
940402	AE2-024 E OI	108.45
940411	AE2-025 C OI	11.69
940412	AE2-025 E OI	54.74

<i>942101</i>	<i>AE2-222 C OI</i>	<i>4.02</i>
<i>942102</i>	<i>AE2-222 E OI</i>	<i>10.09</i>
<i>942381</i>	<i>AE2-251 C OI</i>	<i>16.83</i>
<i>942382</i>	<i>AE2-251 E OI</i>	<i>43.06</i>
<i>918452</i>	<i>AA1-060 E</i>	<i>2.08</i>
<i>919662</i>	<i>AA2-048 E</i>	<i>1.38</i>

16.8 Index 2

(JCPL - JCPL) The 28RED OAKB-28RAR RVR 230 kV line (from bus 206315 to bus 206305 ckt 1) loads from 98.95% to 106.66% (AC power flow) of its emergency rating (869 MVA) for the single line contingency outage of 'JC-P1-2-JCC-230-028'. This project contributes approximately 65.98 MW to the thermal violation.

CONTINGENCY 'JC-P1-2-JCC-230-028'

/* RARITAN RIVER - RED OAK - SOUTH RIVER

JCT(T1034) 230 KV

DISCONNECT BRANCH FROM BUS 206305 TO BUS 206314 CKT 1 /* 28RAR RVR 230.00 28RED OAKA 230.00

DISCONNECT BRANCH FROM BUS 206314 TO BUS 206411 CKT 1 /* 28RED OAKA 230.00 28R11RINGA 230.00 .

DISCONNECT BUS 206314 /* 28RED OAKA 230.00

DISCONNECT BUS 206363 /* 28RDOAKCT2 18.00

DISCONNECT BUS 206364 /* 28RDOAKCT3 18.00

END

<i>Bus Number</i>	<i>Bus Name</i>	<i>Full Contribution</i>
937512	AD2-210 E	0.13
938781	AE1-104 C OI	5.
938782	AE1-104 E OI	12.78
939981	AE1-238 C	53.44
939982	AE1-238 E	140.36
940161	AE2-000 C OI	28.21
940162	AE2-000 E OI	72.19
940361	AE2-020 C OI	5.24
940362	AE2-020 E OI	24.55
940371	AE2-021 C OI	5.24
940372	AE2-021 E OI	24.55
940381	AE2-022 C OI	2.6
940382	AE2-022 E OI	12.18
940401	AE2-024 C OI	23.01
940402	AE2-024 E OI	107.71
940411	AE2-025 C OI	11.61

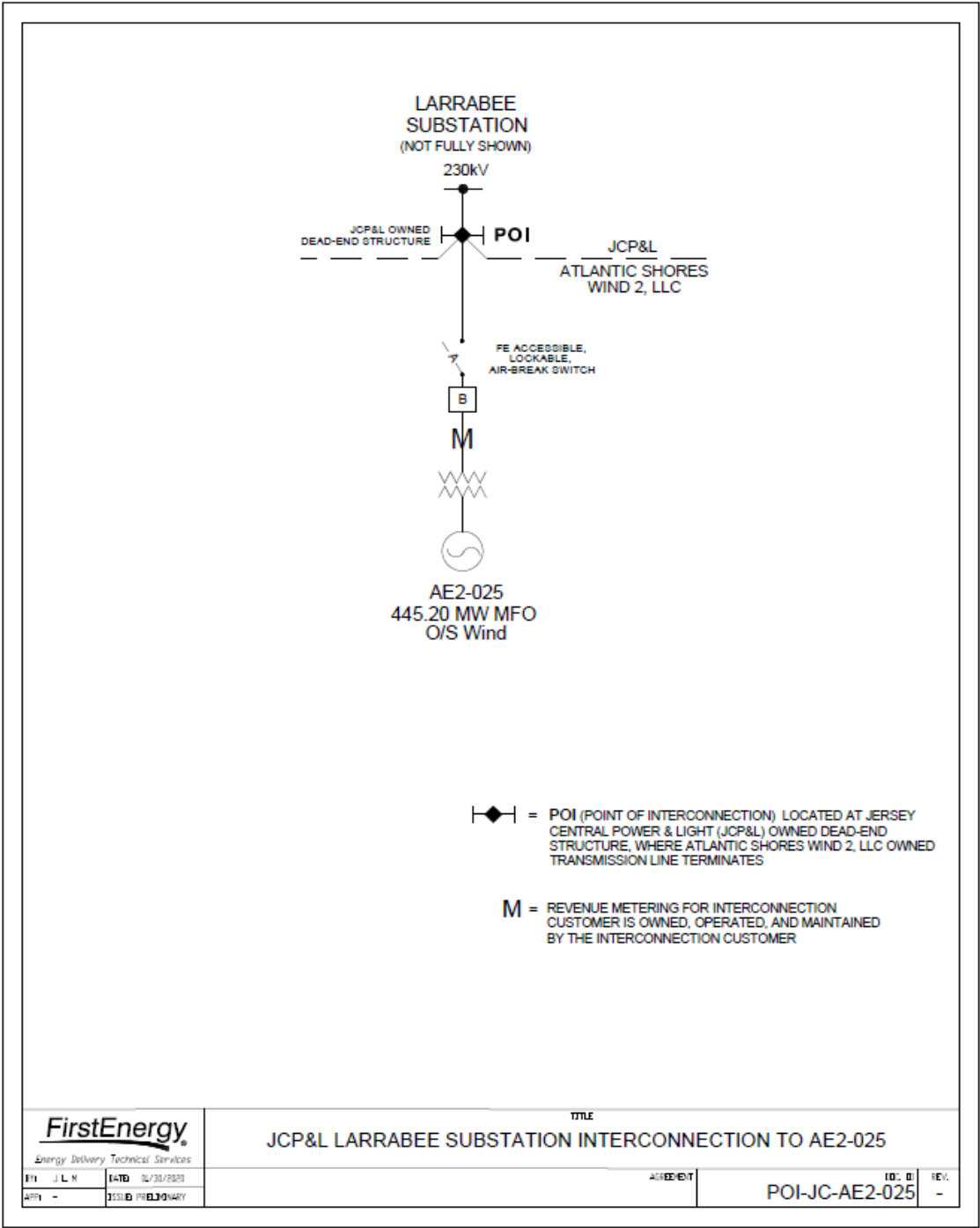
<i>940412</i>	<i>AE2-025 E OI</i>	<i>54.37</i>
<i>942101</i>	<i>AE2-222 C OI</i>	<i>3.97</i>
<i>942102</i>	<i>AE2-222 E OI</i>	<i>9.96</i>
<i>942253</i>	<i>AE2-237 BAT</i>	<i>5.38</i>
<i>942381</i>	<i>AE2-251 C OI</i>	<i>16.61</i>
<i>942382</i>	<i>AE2-251 E OI</i>	<i>42.51</i>
<i>918452</i>	<i>AA1-060 E</i>	<i>2.06</i>
<i>919662</i>	<i>AA2-048 E</i>	<i>1.37</i>

16.10 Contingency Descriptions

Contingency Name	Description
JC-P1-2-JCC-230-027	CONTINGENCY 'JC-P1-2-JCC-230-027' /* RARITAN RIVER - RED OAK - R11 (G1047) 230 KV LINE DISCONNECT BRANCH FROM BUS 206305 TO BUS 206315 CKT 1 DISCONNECT BRANCH FROM BUS 206315 TO BUS 206410 CKT 1 DISCONNECT BUS 206315 DISCONNECT BUS 206362 DISCONNECT BUS 206365 END
JC-P1-2-JCC-230-028	CONTINGENCY 'JC-P1-2-JCC-230-028' /* RARITAN RIVER - RED OAK - SOUTH RIVER JCT(T1034) 230 KV DISCONNECT BRANCH FROM BUS 206305 TO BUS 206314 CKT 1 /* 28RAR RVR 230.00 28RED OAKA 230.00 DISCONNECT BRANCH FROM BUS 206314 TO BUS 206411 CKT 1 /* 28RED OAKA 230.00 28R11RINGA 230.00 . DISCONNECT BUS 206314 /* 28RED OAKA 230.00 DISCONNECT BUS 206363 /* 28RDOAKCT2 18.00 DISCONNECT BUS 206364 /* 28RDOAKCT3 18.00 END
JC-P2-3-JCC-230-026C	CONTINGENCY 'JC-P2-3-JCC-230-026C' /* RARITAN RIVER B145 STUCK BREAKER DISCONNECT BUS 206314 /* RED OAK 230 KV A TAP DISCONNECT BUS 206363 /* RED OAK 230 KV #2 DISCONNECT BUS 206364 /* RED OAK 230 KV #3 DISCONNECT BRANCH FROM BUS 206305 TO BUS 206303 CKT 17 /* RARITAN RIVER #17 230/115 KV TRANSFORMER DISCONNECT BUS 206350 /* SAYREVILLE GENERATOR DISCONNECT BUS 206351 /* SAYREVILLE GENERATOR END

JC-P2-3-JCC-230-26D	CONTINGENCY 'JC-P2-3-JCC-230-26D'	/* RARITAN RIVER B146 STUCK BREAKER
	DISCONNECT BUS 206314	/* RED OAK 230 KV A TAP
	DISCONNECT BUS 206363	/* RED OAK 230 KV #2
	DISCONNECT BUS 206364	/* RED OAK 230 KV #3
	DISCONNECT BRANCH FROM BUS 206305 TO BUS 206303 CKT 13 230/115 KV TRANSFORMER	/* RARITAN RIVER #13
	DISCONNECT BUS 206403	/* WOODBRIDGE GENERATOR
	END	

17 Attachment One: One Line Diagram



AE2-025: LARRABEE 230 KV II

